MIL-H-21291B(SHIPS)
20 May 1969
SUPERSEDING
MIL-H-21291A(SHIPS)
3 February 1961
(See 6.5)

MILITARY SPECIFICATION

HOSE ASSEMBLIES, NONCOLLAPSIBLE, FOR AVIATION FUELS

1. SCOPE

1.1 This specification covers noncollapsible hoses for transporting aviation fuels or any petroleum-base fluid. The hoses are suitable for stowage on a hose reel. The inside diameter of the hoses is 1-1/2 or 2-1/2 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-775 - Hose, Rubber or Fabric (Including Tubing); and Fittings, Nozzles and Strainers, Packaging of.
MIL-G-1149 - Gasket Materials, Synthetic Rubber, 50 and 60, Durometer Hardness.
MIL-P-20693 - Plastic Compounds, Molding and Extrusion, Polyamide.

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods.
FED-STD-162 - Hose, Rubber, Visual Inspection Guide for
FED-STD-601 - Rubber, Sampling and Testing.

PUBLICATIONS

MIL-HANDBOOK-H1 - Cross-Index of Chemically Equivalent Specifications and Identification Code (Ferrous and Nonferrous Alloys).

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual

(Application for copies should be addressed to the American Iron and Steel Institute, 350 Fifth Avenue, New York, New York 10001.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B-21 - Naval Brass Rod, Bar and Shapes.
B-30 - Copper-Base Alloys in Ingot Form for Sand Castings.
B-61 - Steam or Valve Bronze Castings.
B-121 - Leaded Brass Plate, Sheet, Strip and Rolled Bar.
B-143 - Tin Bronze and Leaded Tin Bronze Sand Castings.
B-176 - Brass Die Castings.
D-156 - Saybolt Color of Petroleum Products.
D-380 - Rubber Hose, Testing.
D-381 - Existent Gum in Fuels by Jet Evaporation.
D-2276 - Particulate Contaminant in Aviation Turbine Fuels.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

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NATIONAL BUREAU OF STANDARDS (NBS)
Handbook H-28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20401).

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Construction.

3.1.1 Hose. The hose shall be an assembly of the following components, starting with the inner component:

(a) An inner tube of rubber.
(b) One or more plies of breaker fabric impregnated with rubber.
(c) One or more braids or spiral wraps impregnated with rubber. The braids or spiral wraps may be made from cotton, a synthetic fiber or steel.
(d) A helix of round wire thoroughly imbedded in rubber. The rubber shall be of such thickness that the helix is not contacted by either the inner or outer braids or spiral wraps.
(e) One or more braids or spiral wraps impregnated with rubber. The braids or spiral wraps may be made from cotton, a synthetic fiber or steel.
(f) An outer cover of rubber.

Suggested formulations for these components are given in 6.2.

3.1.2 Couplings. Each length of hose shall have one male coupling and one female swivel coupling attached thereto. Each coupling shall be of the replaceable compression type and shall have an internal tailpiece and a long external tapered sleeve to compress a brass wire helix securely against the hose. The coupling shall be compact and shall not have any sharp protruding edges which could damage the hose cover while stowed on the hose reel. The couplings shall be as shown on figures 1, 2 and 3. The male coupling shall terminate in male tapered pipe threads as shown on figure 2, and the female swivel couplings shall be as shown on figure 3. Threads shall be in accordance with Handbook H-28.

3.1.2.1 Control cable. The material for the control cable shall be commercial corrosion-resisting steel. The cable shall be nylon covered. The bare control cable shall be approximately 1/16 inch outside diameter, and the cable when covered with nylon shall be 1/8 inch outside diameter. The cable shall have 7 strands with 7 wires per strand. The diameter of the wire shall be approximately 0.0075 inch.

3.1.3 Dimensions and weight. The dimensions and weight of the hose shall be as specified in table I.

Table I - Dimensions and weight

Hose	Nominal inside diameter of hose		Test procedure
	1-1/2 inches	2-1/2 inches	
Inside diameter, inches	1-1/2, plus or minus 1/16	2-1/2, plus or minus 1/16	4.4.1
Outside diameter, inches	2-1/8, plus or minus 1/16	3-1/8, plus or minus 1/16	4.4.1
Tube thickness, minimum, inch	0.063	0.063	4.4.1

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Table I - Dimensions and weight (cont'd.)

Hose	Nominal inside diameter of hose		Test procedure
Cover thickness, minimum, inch	0.094	0.094	4.4.1
Length[1/], feet	50, plus or minus 1	50, plus or minus 1	4.4.1
Weight[2/], maximum, pounds	85	125	4.4.2

1/ The hose shall be furnished in 50 foot lengths with couplings unless otherwise specified in the contract or order. In any event, the couplings are not to be considered as part of the hose length.

2/ This is the maximum weight of a 50-foot length of hose complete with couplings.

3.1.4 The hose shall be furnished complete with control cable and contact assemblies as shown on figures 1 to 5, inclusive.

3.2 Branding. Each 50-foot length of hose shall have two brands inlaid in the cover and vulcanized thereto. Brands shall be approximately 4 feet from each end. Letters shall be at least 1/4 inch high. Each brand shall have the following information:

(a) MIL-H-21291.
(b) Manufacturer's name or trade-mark.
(c) Month and year of manufacture.
(d) Size (either 1-1/2 inches or 2-1/2 inches, as applicable).
(e) "AVFUEL S&D"
(f) "150 psi max. W.P."
(g) "22 inches Hg. max. vac."

3.3 Strength and flexibility.

3.3.1 Resistance to vacuum.

3.3.1.1 Initial. The hose shall not collapse or otherwise deform more than 15 percent of its original diameter when subjected to the vacuum test specified in 4.4.3.1.

3.3.1.2 After holding simulated gasoline. The hose section after being filled with simulated aromatic gasoline shall not collapse or otherwise deform more than 15 percent of its original diameter when subjected to the vacuum test specified in 4.4.3.2. When dissected after this test, the hose shall show no evidence of ply separation, blistering, collapse or other damage.

3.3.2 Resistance to proof pressure. When subjected to the proof pressure test specified in 4.4.4, the hose shall not exhibit any of the following:

(a) A change in length of more than 5 percent of the original length.
(b) A change in circumference of more than 6 percent of the original circumference.
(c) Leakage or other signs of weakness.
(d) Electrical contact between the coupling and the intercontrol cable.

3.3.3 Resistance to burst. The hose when subjected to the burst test, 4.4.5, shall withstand 600 p.s.i.g. pressure without leakage if 1-1/2 inches inside diameter, or 500 p.s.i.g. pressure without leakage if 2-1/2 inches inside diameter.

3.3.4 Low temperature flexibility. The tube and cover of the hose shall show no signs of cracking after the hose is subjected to the test specified in 4.4.6.

3.4 Cleanliness. The interior surface of the hose shall be free from particulate materials such as soapstone, talc or graphite to the extent that the weight of sediment contained in the flushing fluid from the tube shall be no more than 10 milligrams per liter when tested as specified in 4.4.7.

3.5 Materials.

3.5.1 Rubber components. The rubber components of the hose shall conform to the requirements shown in table II when tested as specified in 4.4.8.

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Table II - Requirements for rubber components

Properties	Requirements			Test paragraph
	Tube	Cover	Friction compound on reinforcement plies	
Initial properties:				4.4.8.1
Tensile strength, minimum, psi	2600	1800		4.4.8.1.1
Ultimate elongation, minimum, percent	300	300		4.4.8.1.1
Adhesion, minimum, pounds per inch width:				4.4.8.2
Between tube and breaker ply	15			4.4.8.2.1
Between cover and reinforcement ply		15		4.4.8.2.1
Between reinforcement plies and filler on wire helix			15	4.4.8.2.1
Properties after immersion in medium No. 6:				
Tensile strength, minimum, psi	1400	600		4.4.8.1.2
Ultimate elongation, minimum, percent	200	150		4.4.8.1.2
Adhesion, minimum, pounds per inch width:				4.4.8.2
Between tube and breaker ply	9			4.4.8.2.2
Between cover and reinforcement ply		9		4.4.8.2.2
Volume increase, maximum, percent (no shrinkage allowed)	30	100	100	4.4.8.3
Properties after oven aging:				4.4.8.1.3
Tensile strength, minimum, psi	2000	1350		4.4.8.1.1
Ultimate elongation, minimum, percent	250	220		4.4.8.1.1
Resistance to ozone of hose cover		No cracks		4.4.8.4
Nonvolatile extractable material in hose tube, maximum, percent	2.0			4.4.8.5
Saybolt color number of test fluid in hose tube, minimum	+15			4.4.8.6
Phosphate plasticizer in hose tube, maximum, percent	0.0			4.4.8.7

3.5.2 Reinforcement and wire helix.

3.5.2.1 Wire reinforcement. The wire used for reinforcement shall be of high strength corrosion-resisting steel or high strength steel of a good commercial quality. The carbon steel wire shall be galvanized, then copper or brass coated.

3.5.2.2 Fabric reinforcement. The yarn used in reinforcement fabric may be of cotton, nylon, polyester, high tenacity rayon, or other synthetic fiber specifically approved by the Naval Ship Engineering Center (Auxiliary Equipment Branch). The fabric shall be of high quality, free of defects and irregularities and shall be well impregnated with friction compound.

3.5.2.3 Wire helix. The wire in the helix shall be heavily galvanized or copper plated steel with a diameter of not less than 0.035 inch. The cross sectional area of wire per linear inch of longitudinal section on one side of the hose shall be not less than 0.09 square inch for the 1-1/2 inch I.D. hose, and not less than 0.11 square inch for the 2-1/2 inch I.D. hose. The tensile strength of the wire shall be not less than 200,000 pounds per square inch when tested as specified in 4.4.9.1. The wire when wrapped as specified in 4.4.9.2 shall show no defects in the steel. Flaking or peeling of the galvanized or copper coating on the wire shall not be considered a defect.

3.5.3 Couplings, fittings and control cable. The materials of the couplings and gasket for the swivel coupling shall be as specified in table III. Parts may be made of other materials having equivalent physical properties and chemical compositions and which will perform the intended function. Acceptable equivalent materials shall be those materials of the same form (castings, bar, forging and similar forms) as listed in the same code number group of Handbook MIL-HANDBOOK-H1.

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Table III - Material for couplings, fittings and contact assembly

Piece Number	Name of piece	Material	Applicable document
1	Button	Brass	ASTM-B-121
2	Spring	Corrosion-resisting steel	AISI, type 302
3	Front nut	Brass	ASTM-B-121
4	Spider	Molded nylon	MIL-P-20693, type I
5	Contact rod	Brass	ASTM-B-121
6	Washer	Brass	ASTM-B-121
7	Main body	Brass	ASTM-B-121
8	Lead washer	Virgin lead	
9	Contact shaft	Brass	ASTM-B-121
10	Cable connector	Brass	ASTM-B-121
11	Hose assembly	As required (see 3.5)	As required
12	Coupling, hose	Brass, castings	ASTM-B-30; Alloy 6C or ASTM-B-176; Alloy ZS33lA
		Brass, naval	ASTM-B-121, Alloy B
		Brass, commercial	ASTM-B-121
		Bronze, valve	ASTM-B-61 or B-143 alloy 2A
13	Gasket	Chrome tanned, leather impregnated with thiokol	
		Rubber sheet	MIL-R-1149, type II, class 5

3.6 Workmanship. Workmanship shall be first class in every respect. Broken, spliced or omitted reinforcing wires shall be avoided. The defects of the cover and tube shown in FED-STD-162 shall be avoided.

3.6.1 The occurrence of broken, spliced or omitted reinforcing wires shall be noted when preparing specimens for the tests specified in 4.4.3.2, 4.4.6, and 4.4.8.

3.7 There shall be no deviations from the requirements of this specification unless prior approval has been obtained in writing from the Naval Ship Engineering Center (NAVSEC), Auxiliary Equipment Branch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 Lot. All hose assemblies offered for delivery at one time, and not to exceed 1200 lengths of hose, shall be considered a lot.

4.2.2 Sampling for visual and dimensional examination. Samples of hose assemblies shall be taken at random from each lot in accordance with table IV for the inspection specified in 4.3.1. If in any sample the number of defective hose assemblies exceeds the acceptance number for that sample, the lot represented by the sample shall be cause for rejection.

Table IV - Sampling for visual and dimensional examination

Number of hose assemblies in lot	Number of hose assemblies in sample	Major defectives AQL = 1.5 percent		Total defectives AQL = 6.5 percent	
		Acceptance number	Rejection number	Acceptance number	Rejection number
8 and under	all	0	1	1	2
9 to 90	8	0	1	1	2
91 to 280	32	1	2	5	6
281 to 500	50	2	3	7	8
501 to 1200	80	3	4	10	11

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4.2.3 Sampling for vacuum and proof tests. Samples of hose assemblies shall be taken at random from each lot in accordance with table V for the tests specified in 4.4.3 and 4.4.4.

Table V - Sampling for vacuum and proof tests

Number of hose assemblies in lot	Number of hose assemblies in sample	AQL = 1.5 percent	
		Acceptance number (test failures)	Rejection number (test failures)
8 and under	all	0	1
9 to 90	8	0	1
91 to 280	32	1	2
281 to 500	50	2	3
501 to 1200	80	3	4

4.2.4 Sampling for all other tests. With each 100 hose assemblies or less offered for quality conformance inspection, the following samples shall be furnished:

(a) One hose assembly shall be provided in addition to the quantity required for delivery. This hose assembly shall be taken at random for the tests specified in 4.4.3.2, 4.4.5, 4.4.6, 4.4.7 and 4.4.8.
(b) Vulcanized samples, 6 by 6 inches by 0.080 ± 0.005 inch in thickness of the friction and filler compounds used to impregnate the reinforcement and surround the wire helix, or both, shall be provided for the test specified in 4.4.8.3. These samples shall be accompanied by a written certification from the manufacturer that the samples have the same compositions as the friction and filler, respectively, used in the hose and have equivalent cures.
(c) A straight 6 foot length of the helix wire used in the hose offered for quality conformance inspection for the tests specified in 4.4.9.

4.2.5 Action in case of nonconformance. If any one of the samples is found not to conform with the requirements of this specification, the lot which it represents shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Visual and dimensional examination. Each of the sample hose assemblies selected in accordance with 4.2.2 shall be examined to verify conformance with the requirements of this specification regarding all parts of the hose assembly which do not involve tests. Visual defects shall be determined and evaluated in accordance with FED-STD-162. Length, outside diameter and inside diameter shall be determined as specified in 4.4.1. Weight shall be determined as specified in 4.4.2.

4.3.1.1 Definition of major defective. The following defines the term "major defective" used in table IV. Any hose assembly in the sample containing one or more major visual defects, one or more dimensional defects, or five or more minor visual defects shall be considered a major defective. For lengths of hose other than 50 feet, the number of minor defects constituting a major defective shall be proportional to the length.

4.3.1.2 Definition of minor defective. Any hose assembly containing one to four minor defects shall be considered a minor defective, and shall be included in the count for total defectives in table IV.

4.3.1.3 Rejection. Any hose assembly rated as a major defective according to 4.3.1.1 shall be rejected. Should the number of defective lengths of hose in any sample exceed the acceptance number for total defectives for that sample, the lot represented by the sample shall be rejected, as specified in table IV.

4.4 Test methods.

4.4.1 Dimensions of hose. The procedure given in ASTM D380 shall be used to determine the inside diameter, outside diameter, tube thickness, cover thickness and length of the hose. Inside diameter, outside diameter and length shall be measured on each of the sample hose assemblies selected in accordance with 4.2.2. Tube thickness and cover thickness shall be measured at six points equally spaced along the length of the sample hose selected in accordance with 4.2.4(a). Each point shall be spaced 60 degrees apart around the circumference. The thickness values found shall not be averaged. Each thickness value must conform to the requirements specified in table I.

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4.4.2 Weight of hose. The weight of each sample hose assembly, complete with couplings, selected in accordance with 4.2.2 shall be determined with a precision of plus or minus 0.5 percent.

4.4.3 Resistance to vacuum.

4.4.3.1 Initial. Each of the sample hose assemblies selected in accordance with 4.2.3 shall be subjected to a vacuum equivalent to a 22 inch height of mercury for a period of not less than 5 minutes. The decreases in diameter of the hose where deformation occurs shall be measured in accordance with ASTM D380.

4.4.3.2 After holding simulated gasoline. A 5 foot length shall be cut from the sample hose assembly submitted in accordance with 4.2.4(a). One end shall be plugged and the hose filled with medium number 6 of method 6001 of FED-STD-601. The other end shall then be closed and the hose let stand for 46 hours at a temperature of 70° to 80°F. At the end of this period, the fluid shall be drained from the hose. Within 5 minutes after removal of the fluid, the hose length shall be bent to a radius of 15 inches for the 1-1/2 inch I.D. hose, or 18 inches for the 2-1/2 inch I.D. hose. The bent hose length shall then be subjected to a vacuum equivalent to a 22 inch height of mercury for a period of not less than 15 minutes. The decrease in diameter of the hose where deformation occurs shall be measured. After release of the vacuum, the hose shall be dissected longitudinally and examined for evidence of ply separation, blistering, collapse or other damage.

4.4.4 Resistance to proof pressure. Each of the sample lengths of hose selected in accordance with 4.2.3 and previously used for the vacuum test, 4.4.3.1, shall be measured for length and circumference as specified in ASTM D380. Each hose shall then be subjected to a hydrostatic pressure of 200 p.s.i.g. for a period of not less than 10 minutes. At the end of this period while the hose is still under pressure, its length and circumference shall again be measured, it shall be examined for leakage or other signs of weakness, and the contact assembly shall be checked electrically to assure that there is no electrical contact between the couplings and intercontrol cable.

4.4.5 Resistance to burst. A 5 foot length shall be cut from the sample hose selected in accordance with 4.2.4(a). Couplings shall be attached to the 5 foot length and the ends blanked off. The hose shall then be subjected to increasing hydrostatic pressure with the rate of pressure rise not exceeding 100 p.s.i. per minute. The pressure at which leakage occurs shall be noted.

4.4.6 Low temperature flexibility. A 4 foot length shall be cut from the sample hose submitted in accordance with 4.2.4(a). One end shall be plugged and the length filled with medium number 6 of method 6001 of FED-STD-601. The other end of the hose shall then be plugged, and the hose exposed to a temperature of minus 22° + 2°F for 24 hours while held straight. At the end of this period and while at minus 22°F, the hose shall be bent 90° around a mandrel 30 inches in diameter within 4 seconds. The fluid shall be drained from the hose, and the hose cut in half longitudinally. The tube and cover shall be examined for cracks or other damage.

4.4.7 Cleanliness test.

4.4.7.1 Specimen. A 4 foot length shall be cut from the sample hose submitted in accordance with 4.2.4(a). Care shall be taken that no extraneous dust or other contaminant enters the hose tube. The hose shall be closed with clean plugs at each end. Each plug shall have an outlet hole connected via a pipe nipple to a valve as shown on figure 6. The interiors of the pipe nipples and valves shall be carefully cleaned prior to use.

4.4.7.2 Apparatus. A machine is required that will impart a rotary motion in a vertical plane to a rigid bar that is attached at its center to a horizontal axle driven at 12 + 1 RPM. Four adjustable band clamps shall be attached symmetrically to the rigid bar to hold the specimen firmly. Such an apparatus is shown on figure 6.

4.4.7.3 Procedure. The specimen, having been sealed at one end by a prepared plug, shall be filled with medium number 6 of method 6001 of FED-STD-601 and hung vertically for 16-24 hours. The fluid shall then be drained from the specimen and 2 liters of fresh medium number 6 added, and the open end plugged. The specimen shall then be clamped securely to the rotating bar so that the ends are spaced evenly from its center. After 120 minutes of rotary motion at 12 RPM and 70° to 80°F, the first liter of the fluid shall be withdrawn for analysis. The amount of sediment shall be determined in milligrams per liter in accordance with ASTM method D2276 using an 0.8 micron pore filter. The tests shall be performed in duplicate and the results averaged. A control analysis shall determine the amount of contaminant present per liter in the medium number 6 used for the test. The average amount of

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sediment found in the specimen shall be reduced by the amount found present in the control, and the difference recorded as the amount of particulate contaminate in the hose.

4.4.8 Tests performed on rubber components. All tests except the test for volume increase of the friction compound after immersion in test fluid shall be performed on specimens prepared from the sample hose selected in accordance with 4.2.4(a) to determine conformance with 3.5.1.

4.4.8.1 Tensile properties. All tests for tensile properties shall be performed on dumbbell specimens cut by die III of method 4111 of FED-STD-601.

4.4.8.1.1 Initial tensile properties. Initial tensile strength and ultimate elongation at break shall be determined by methods 4111 and 4121, respectively, of FED-STD-601.

4.4.8.1.2 Tensile properties after immersion. The tensile strength and ultimate elongation after immersion in medium number 6 of method 6001 of FED-STD-601 shall be determined by method 6111 of FED-STD-601. Tensile strength shall be based on the swollen cross-sectional areas of the specimen.

4.4.8.1.3 Tensile properties after oven aging. The tensile strength and ultimate elongation shall be determined after oven aging for 70 ± 1/4 hours at 212° ± 2°F in accordance with method 7221 of FED-STD-601.

4.4.8.2 Adhesion. The adhesion between the several parts of the hose shall be determined initially and after immersion in test fluid. The adhesion shall be such that the rate of separation will not be greater than 1 inch per minute under the minimum loads shown in table II.

4.4.8.2.1 Initial adhesion. The initial adhesion between the tube and breaker ply, between the cover and outer reinforcement plies, and between the reinforcement plies and the filler on the wire helix shall be determined by method 8021 of FED-STD-601.

4.4.8.2.2 Adhesion after immersion. The adhesion after immersion shall be determined as specified in 4.4.8.2.2.1 through 4.4.8.2.2.3.

4.4.8.2.2.1 Test specimen. A 2-foot length shall be cut from the sample hose obtained as specified in 4.2.4(a).

4.4.8.2.2.2 Apparatus. The apparatus for subjecting the test specimen to the action of the immersion fluids shall be that shown on figure 7, or equivalent.

4.4.8.2.2.3 Procedure. The specimen shall be positioned in the apparatus and filled with medium number 6 of method 6001 of FED-STD-601 to at least the height of the fluid on the outside of the specimen. The fluid surrounding the outside of the specimen shall be medium number 4 of method 6001 of FED-STD-601. The specimen shall remain in contact with the test fluids for a period of 46 + 1/4 hours at a temperature of 70° to 80°F. At the end of this period the specimen shall be removed and dried for a period of time not to exceed 30 minutes. The adhesion test specimens shall be cut from the specimen hose length during the drying period. The adhesion shall be determined within 2 minutes after completion of the drying period in accordance with method 8021 of FED-STD-601.

4.4.8.3 Change in volume. The percent change in volume after immersion shall be determined in accordance with method 6211 of FED-STD-601. The specimen shall be immersed in medium number 6 of method 6001 of FED-STD-601 for 46 ± 1/4 hours at 70° to 80°F.

4.4.8.4 Resistance to ozone.

4.4.8.4.1 Test specimen. The specimen shall be a strip 1 by 5 inches taken lengthwise from the cover of the sample hose obtained as specified in 4.2.4(a). The specimen shall be buffed smooth on both sides. The maximum thickness after buffing shall be 0.100-inch. Duplicate specimens shall be tested.

4.4.8.4.2 Apparatus. The apparatus shall consist of a test chamber with an ozone generating source outside the chamber capable of maintaining an ozone-air atmosphere in the chamber at the temperature concentration desired, a circulating-air oven, as described in method 7221 of FED-STD-601, and adjustable specimen holders similar to the holder shown on figure 7311A of method 7311 of FED-STD-601.

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4.4.8.4.3 Procedure. The specimen shall be clamped firmly at the ends in the holder at 20 percent extension and placed in an oven for 16-24 hours at 100° ± 2°F. After this preliminary exposure period the specimen shall be placed in an ozone chamber at 100° ± 2°F with an ozone-air atmosphere of 1.00 ± 0.05 parts of ozone per one million parts of air by volume. After exposure for 166 ± 1/2 hours the specimen shall be removed from the ozone chamber and examined for surface cracking with a 7X magnifying lense. Cracks in the rubber within 1/8 inch of the clamps shall be ignored since they may be due to excessive strain in this region.

4.4.8.5 Nonvolatile extract.

4.4.8.5.1 Test specimen. The specimen shall consist of strips of the hose tube about 1/16-inch square by 1-1/2 inches long cut from the sample hose of 4.2.4 (a). The total weight of strips shall be about 5 grams. Duplicate specimens shall be tested.

4.4.8.5.2 Procedure. The weighed specimen shall be placed in an extraction thimble then inserted in a syphon cup of an extraction apparatus as shown in method 6001 of FED-STD-601. One hundred milliliters of medium number 4 of method 6001 of FED-STD-601 shall be added to the extraction flask. Extraction shall be carried out for 46 ± 1/2 hours at such a rate that 2-1/2 to 3 minutes are required to fill and empty the syphon cup. The specimen and flask shall be rinsed with 25 milliliters of fresh fluid, and the wash added to the extract. The nonvolatile residue from the extract shall be determined in accordance with ASTM method D381, except that the evaporation time shall be 45 minutes. The weight of the residue shall be expressed as percent of the original weight of the specimen. The results of duplicate tests shall be averaged. The residue shall be saved for the test for phosphate plasticizer, 4.4.8.7.

4.4.8.6 Saybolt color number. The specimen shall consist of a 15-inch length cut from the sample hose obtained as specified in 4.2.4(a). The specimen shall be sealed at one end, filled within 2 inches of the top with medium number 6 of method 6001 of FED-STD-601 and capped to prevent evaporation or adulteration by extraneous material. A suitable arrangement is shown on figure 8. After 168 ± 2 hours the Saybolt color number of the fluid extract shall be determined according to ASTM method D156. Duplicate tests shall be performed and the results averaged. The Saybolt color number of the uncontaminated medium number 6 used shall be plus 25, minimum.

4.4.8.7 Phosphate plasticizer. The presence of phosphate plasticizers in the tube of the hose shall be determined qualitatively in accordance with method 15825 of FED-STD-601, using about 0.2 grams of the nonvolatile residue from 4.4.8.5.2. The procedure shall be repeated if phosphorous is found, omitting addition of the nonvolatile residue in order to determine whether phosphorous was present in any of the reagents used.

4.4.9 Tests performed on helix wire.

4.4.9.1 Tensile strength. The tensile strength of the wire shall be determined in accordance with FED-STD-151.

4.4.9.2 Wrapping test. Each of the samples of helix wire selected in accordance with 4.2.4(c) shall be wrapped six full turns around a mandrel having a diameter three times the diameter of the wire. The wrapped wire shall be examined for defects.

4.5 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to determine conformance with the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4)

5.1 Domestic shipment and early equipment installation and for stowage of onboard repair parts.

5.1.1 Preservation and packaging. Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

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5.1.1.1 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, supplier's name and destination.

5.1.2 Onboard repair parts. Onboard repair parts shall be preserved and packaged level A, packed level A or C and marked in accordance with MIL-H-775.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides the various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Hose and hose assemblies.

5.2.1.1.1 Preservation and packaging. Hose and hose assemblies shall be preserved and packaged level A or C in accordance with the requirements of MIL-H-775.

5.2.1.1.2 Packing. Hose and hose assemblies packaged as specified, shall be packed level A, B, or C in accordance with the requirements of MIL-H-775.

5.2.2 Marking. In addition to any special marking required by the contract or order, or herein, interior and exterior shipping containers shall be marked with the date (month and year) of manufacture and in accordance with MIL-STD-129).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Number of lengths of hose assemblies required.
(c) Preservation, packaging, packing and marking requirements other than those required by 5.1 (see 5.2).

6.2 Suggested rubber formulations for the tube, friction, filler and cover of hose are as follows:

Tube stock		Friction stock	
Compounding ingredients 1/	Parts by weight	Compounding ingredients 1/	Parts by weight
Paracril CLY	100	Hycar 1041	100
Philblack N330	60	Philblack N550	10
Sulfur	1.5	Hi Sil 233	30
Protox 166	5	Dixie clay	30
Stearic acid	1	Sulfur	2
Hycar 1312	10	Protox 166	5
TP90B	10	Stearic acid	1
Altax	1	Cumar P10	30
Methyl Tuads	0.2	Bardol B	15
		Hylene MP	10
		Altax	2

See footnote at end of table.

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Cover and filler stock	
Compounding ingredients[1]	Parts by weight
Paracril OZO	100
Regal 300	30
Philblack N330	30
Sulfur	1.5
Protox 166	5
Stearic acid	1
Flexzone 6H	3
Cumar P25	15
Plasticizer SC	10
Flexol DOP	10
Altax	1.5
Thionex	0.2

Cement	
Compounding ingredients[1]	Parts by weight
Tube stock	25
Methyl isobutyl ketone	100
Chemlok 205	135

Volume of Chemlok 205 equal to combined volumes of tube stock and methyl isobutyl ketone.

[1] These were the specific ingredients used when developing these formulations. Many of these ingredients are available under other brand names. It is not the intention of this listing to limit the choice of the commercial source of an ingredient or to infer that one brand is better than another.

6.3 The furnishing of the formulations in 6.2 solely for the fabrication of hose for Naval use. The use of this information does not constitute any agreement or obligation by the Department of the Navy to procure hoses made of these formulations. Also, the use of this information does not guarantee compliance with this specification, nor will it relieve the manufacturer from having the hoses tested against the applicable test requirements of this specification.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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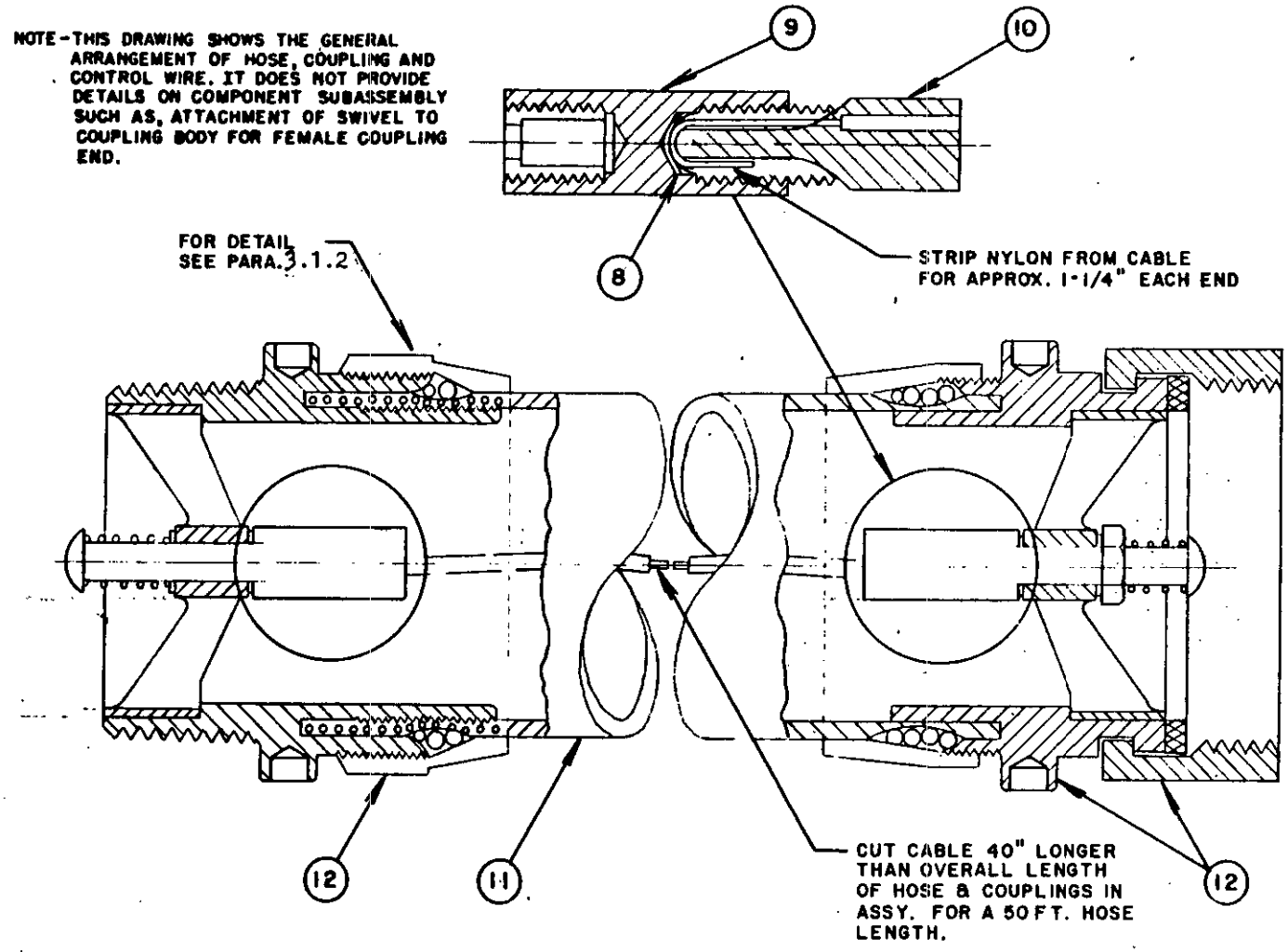
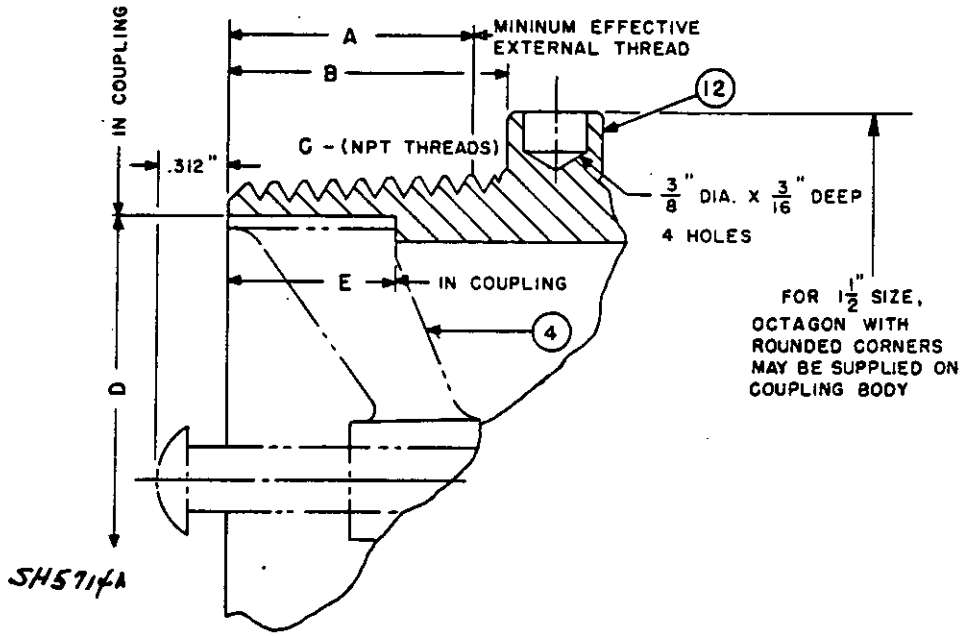


FIGURE 1 - HOSE AND CONTROL WIRE ASSEMBLY

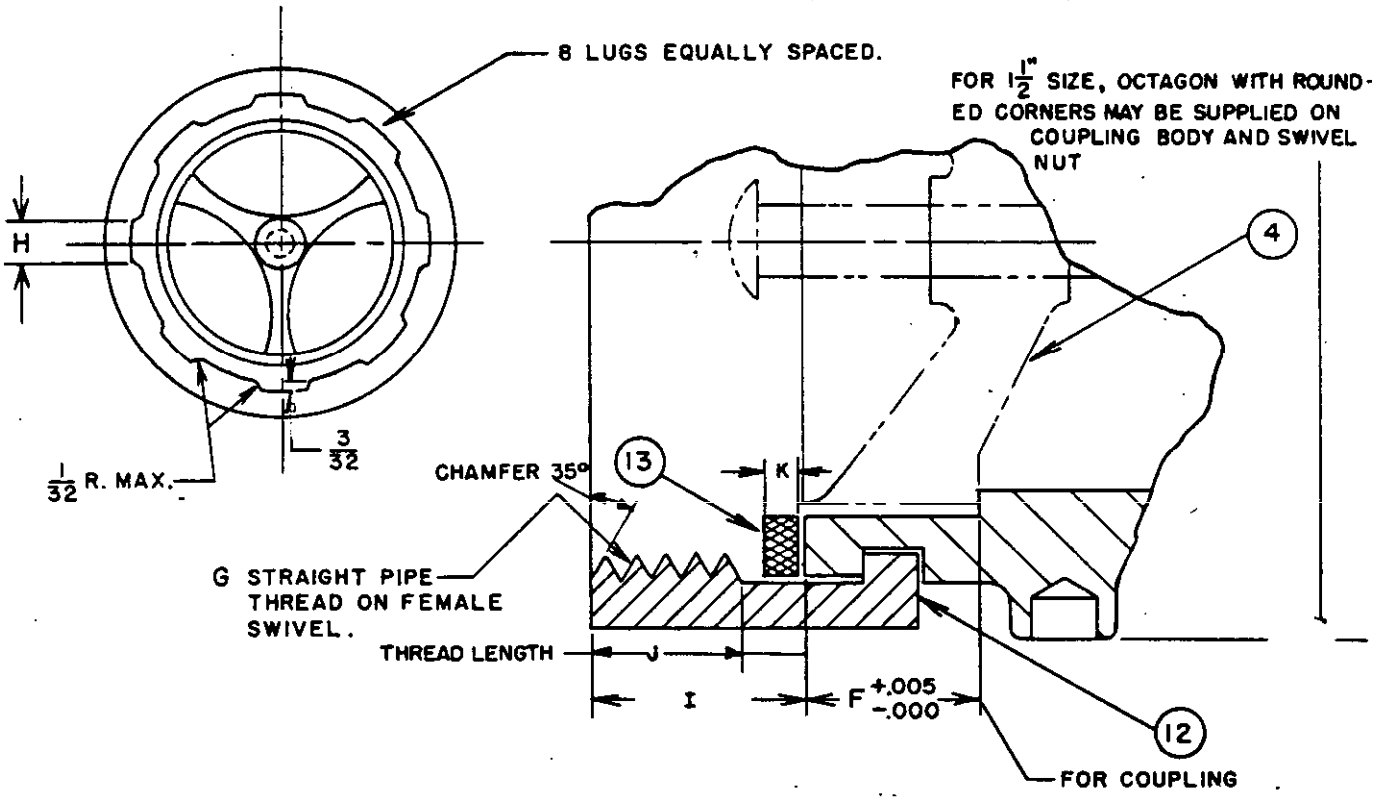
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Size	A	B Minimum	C	D	E
1-1/2	0.7235	15/16	1-1/2 - 11-1/2 NPT	1.520 ± .005	0.560 +.005 -.000
2-1/2	1.1375	1-1/4	2-1/2 - 8 NPT	2.520 ±.005	.781 +.005 -.000

Figure 2 - Male coupling dimensions

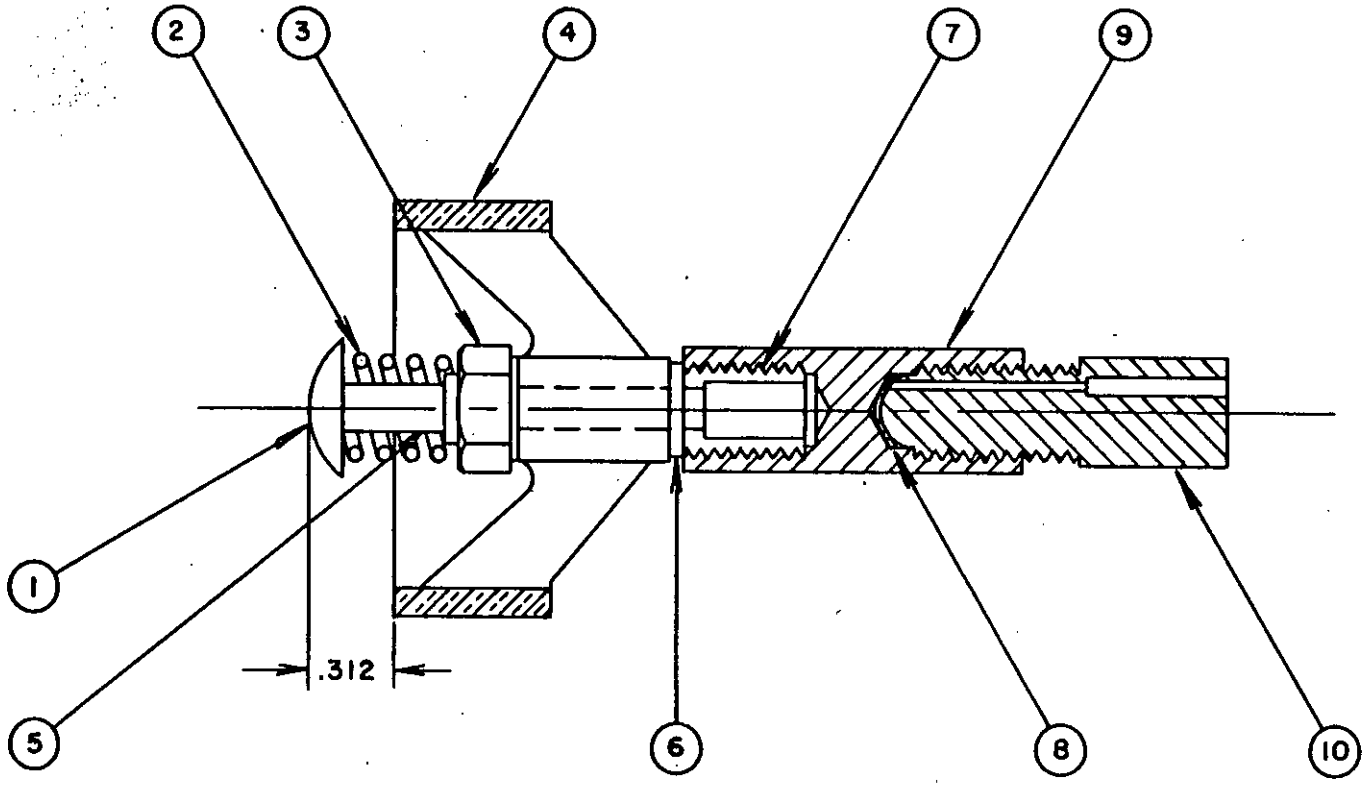
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Size	F	G	H	I	J	K
1-1/2	0.560	1-1/2	1/2	19/32	15/32	1/8
2-1/2	.781	2-1/2	5/8	15/16	11/16	3/16

Figure 3 - Partial detail of female coupling end

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FIGURE 4 - CONTACT ASSEMBLY

(SEE 3.5.3 FOR MATERIALS)

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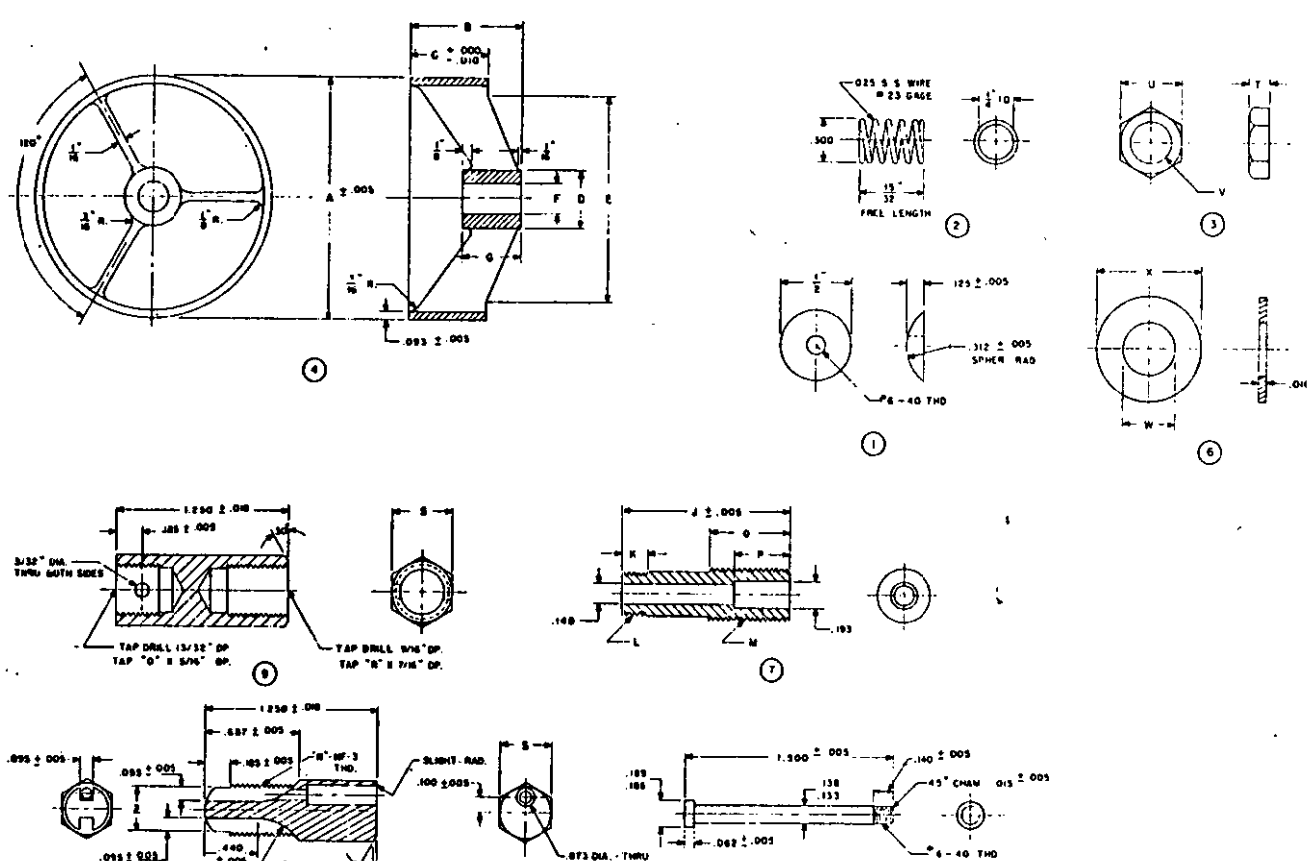


Figure 5 - Details of contact assembly.

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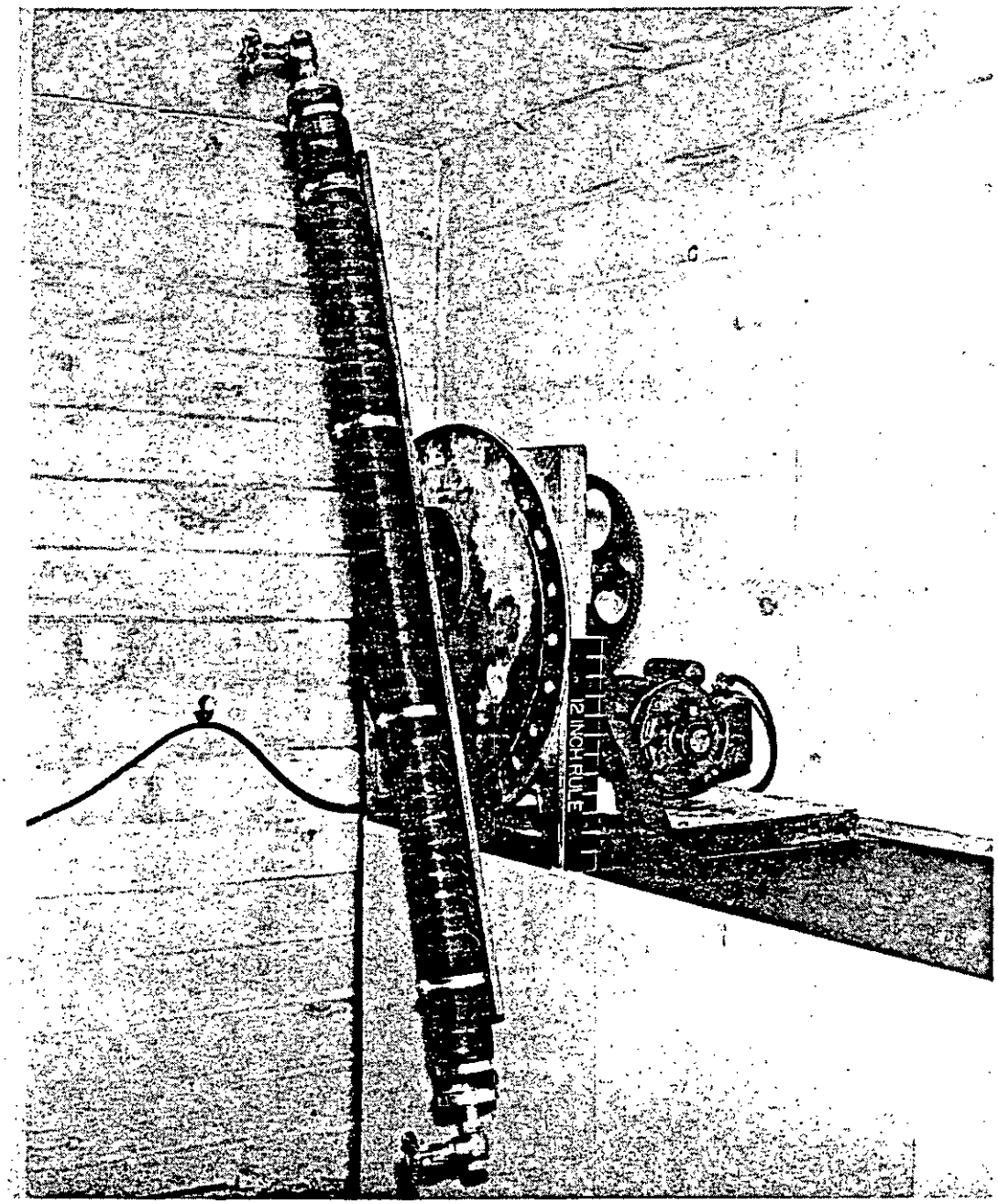


Figure 6 - Rotary apparatus used in sedimentation test of 2-1/2 inch I.D. aircraft fueling hose.

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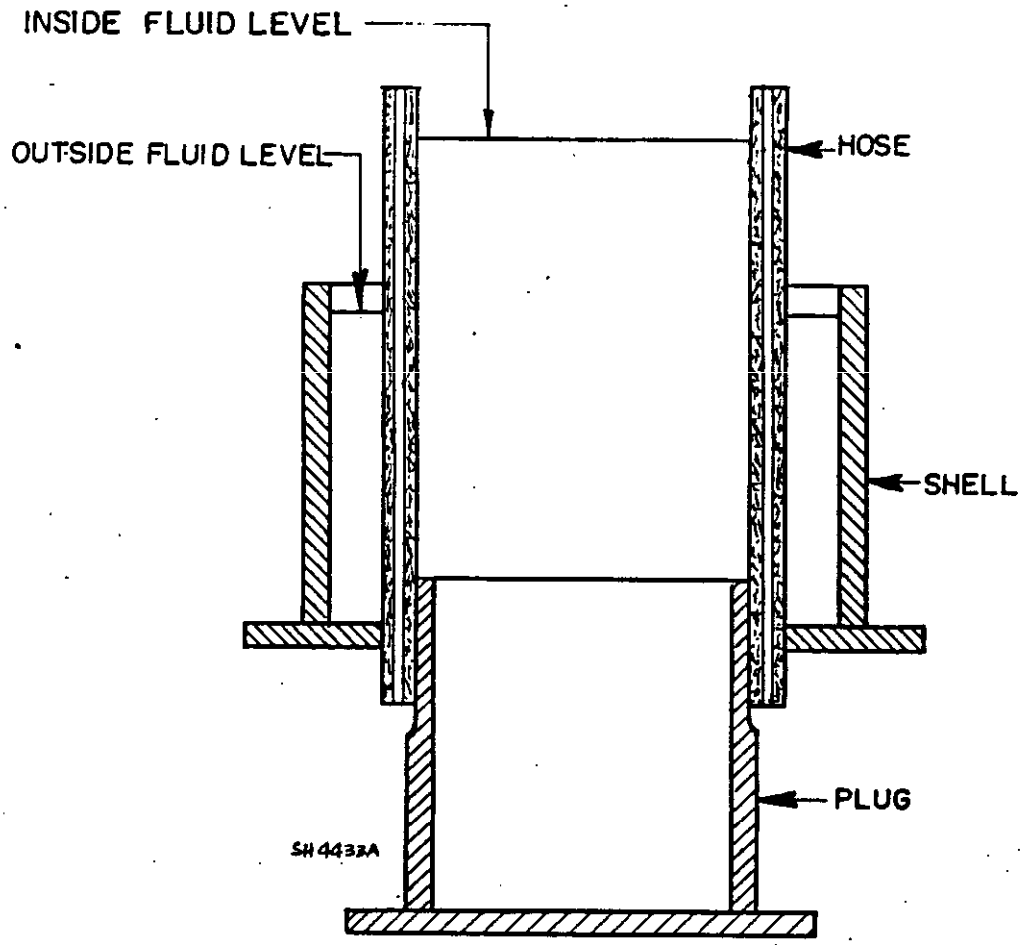


Figure 7 - Jig for conditioning straight section of hose.

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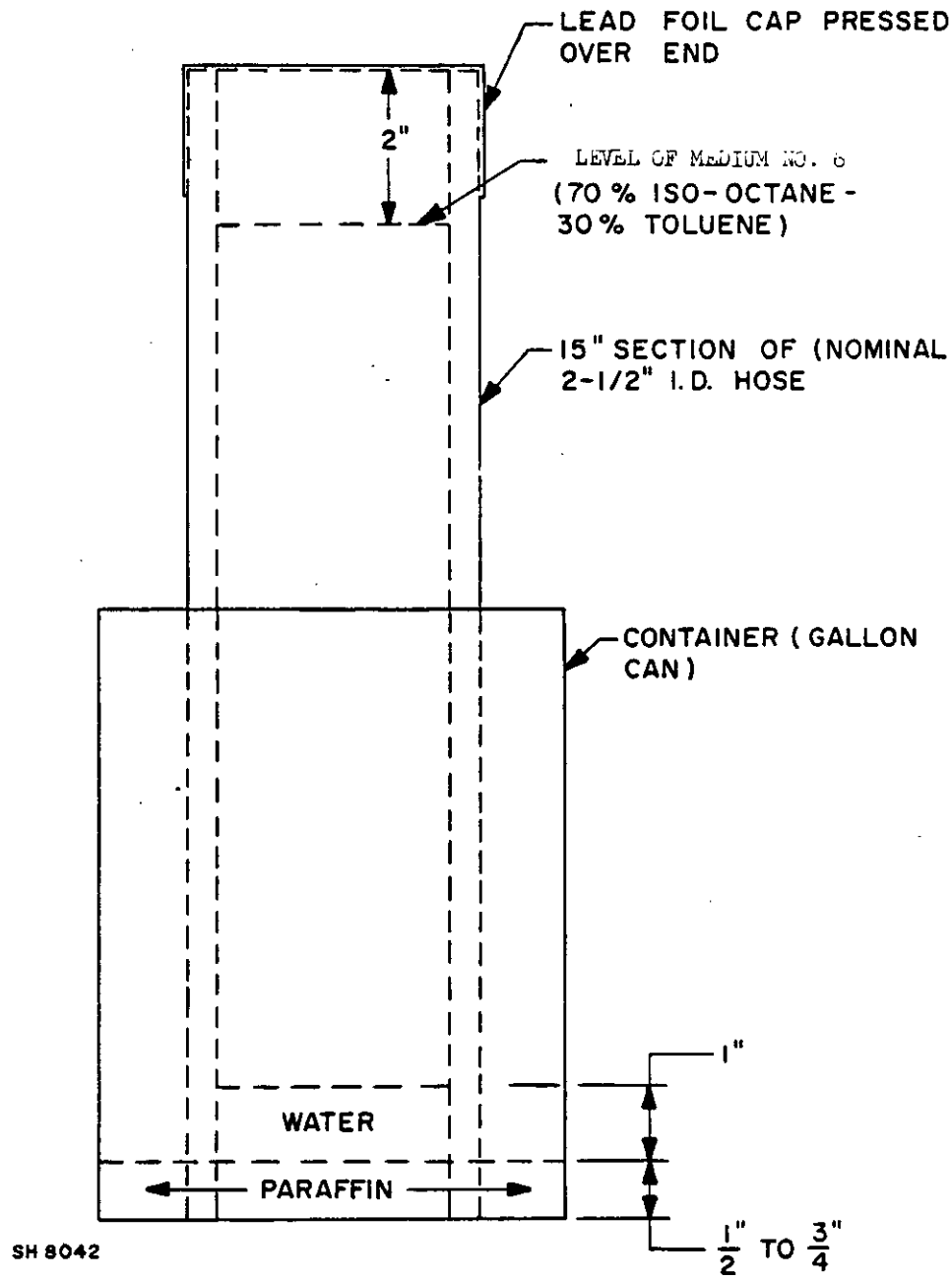


Figure 8 - Assembly for obtaining extract for use in determining Saybolt color number.

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